

Lab 1: Requirements Engineering & UML Use-Case Modelling

Topic: Student Project Portfolio & Showcase App

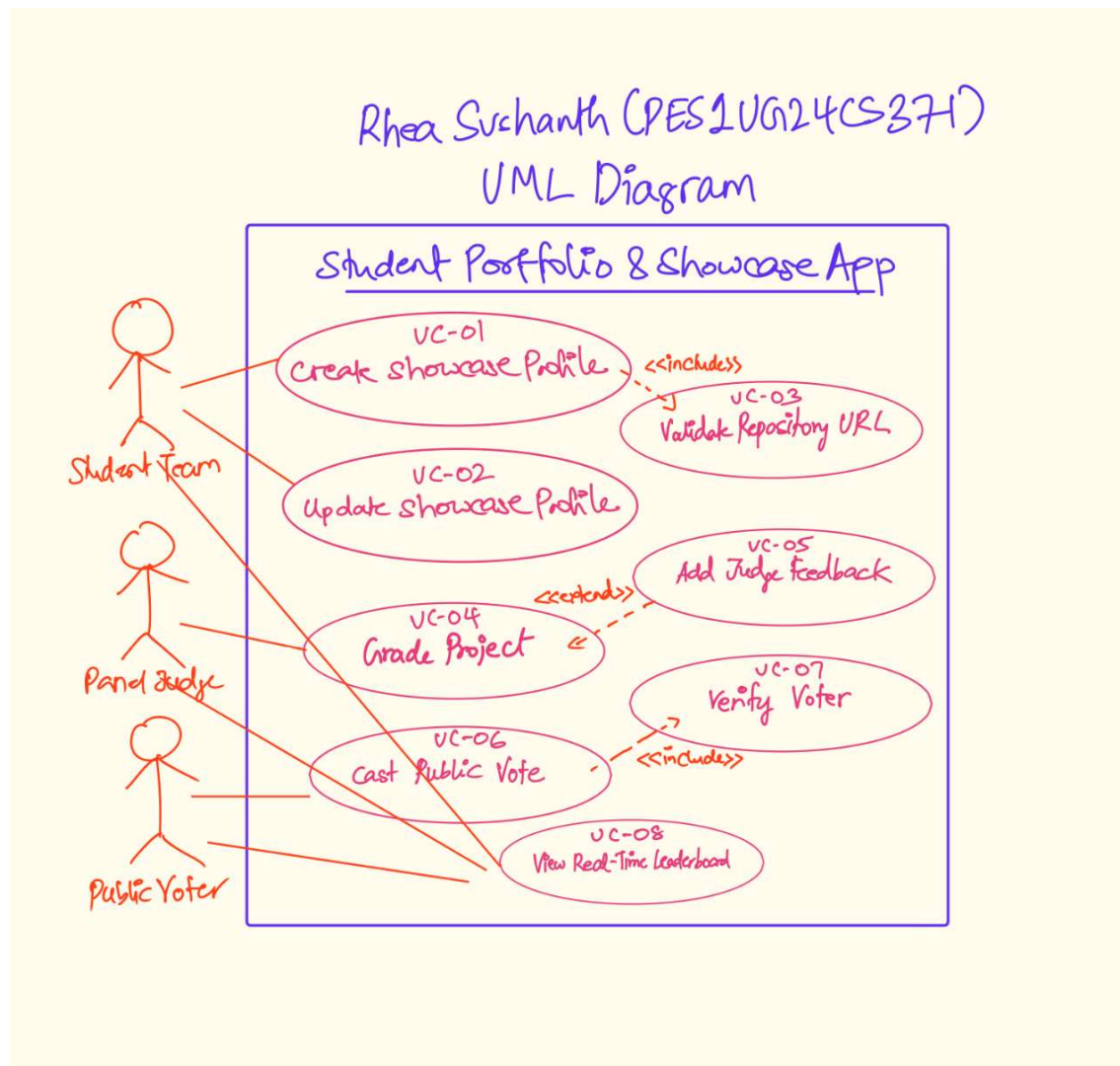
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1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable student teams to create showcase profiles containing abstract summaries, demo video links, and verified GitHub repository URLs.	High	Pass: Profile is published to the showcase catalog upon validation of the repository link. Fail: An invalid URL format is accepted.	Core onboarding functionality; establishes the project record judges and the public will interact with.
FR-002	Functional	The system shall allow a Panel Judge to grade a submitted project using a weighted digital rubric covering at least 4 criteria (e.g., innovation, technical depth, presentation, impact).	High	Pass: A submitted score is saved only when all rubric criteria are rated and the weighted total is computed and displayed to the judge. Fail: A score is saved with one or more empty criteria.	Digital rubric grading is the core judging mechanism specified in the scenario and must produce a consistent, auditable score.
FR-003	Functional	The system shall allow a verified public voter to cast exactly one vote per project per event, rejecting any repeat vote attempt from the same verified identity.	High	Pass: A second vote attempt from the same verified voter on the same project returns a 'duplicate vote' response and the vote count does not increase. Fail: The vote count increases on a repeat attempt.	Directly implements the 'strict anti-sybil protection' requirement stated in the problem context.
FR-004	Functional	The system shall recalculate and display the public leaderboard ranking immediately after each valid vote or judge score is submitted.	Medium	Pass: The leaderboard rank order reflects a newly submitted vote/score within one page refresh with no manual trigger. Fail: The leaderboard requires a manual admin action to update.	Real-time visibility keeps the expo engaging for students and the public and is the functional counterpart to NFR-001's latency target.
FR-005	Functional	The system shall allow a student team to edit their showcase profile (abstract, video link, repository URL) at any time before the submission deadline, after which the profile is locked from further edits.	Medium	Pass: An edit attempt after the configured deadline timestamp is rejected with a 'submissions closed' message and the stored profile is unchanged. Fail: An edit after the deadline is saved.	Prevents last-minute tampering with submissions after judging has begun, preserving fairness of the showcase.
NFR-001	Nonfunctional (Performance & Security)	The public voting and leaderboard calculations shall update in real time with an API response latency under 300 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: p95 latency exceeds 300 ms during the load test.	Keeps the live-expo experience responsive for a large concurrent audience of voters and judges.
NFR-002	Nonfunctional (Security)	The system shall detect and block duplicate or bot-originated votes using device fingerprinting and one-time verification (e.g., OTP/institutional SSO), achieving at least 99% duplicate-vote detection accuracy.	High	Pass: A controlled test batch of 100 duplicate/bot vote attempts results in at least 99 being blocked and logged. Fail: More than 1 duplicate/bot vote in the batch is counted toward the leaderboard.	Provides the measurable, testable backbone for the 'strict anti-sybil protections' stated in the problem context, complementing FR-003.

2. UML Use - Case Diagram



3. Use-Case Flow Specification

UC-06: Cast Public Vote

Primary Actor	Public Voter
Related actors	System (includes UC-07: Verify Voter / Anti-Sybil Check)
Preconditions	1. The project expo's public voting window is open. 2. The voter has an active account/session on the showcase platform (e.g., via institutional SSO or a registered email). 3. The target project's showcase profile has been published (UC-01 completed).
Postconditions	Success: Exactly one vote is recorded against the selected project for this voter, and the public leaderboard reflects the updated vote count within the NFR-001 latency target. Failure: No vote is recorded, and the voter is informed of the reason.

Main Success Scenario – Happy Path

1. Public Voter opens the showcase catalog and browses published project profiles.
2. Voter selects a project and taps “Vote for this project.”
3. System checks the voter's session and triggers UC-07: Verify Voter (Anti-Sybil Check) to confirm the voter has not already voted and is not a bot/duplicate identity.
4. System confirms the voter is unique and eligible for this event.
5. System records the vote against the selected project and timestamps it.
6. System recalculates the leaderboard ranking to include the new vote (per NFR-001, within 300 ms).
7. System displays a confirmation message (“Your vote has been recorded”) and disables further voting on this project for that voter.
8. Use case ends successfully.

Alternate Flow

3a. Duplicate Vote / Anti-Sybil Check Fails

- a) 3a1. System detects that the voter's identity (or device fingerprint) has already voted for this project, or flags the attempt as likely bot/sybil activity.
- b) 3a2. System rejects the vote and displays a message: “A vote from this identity has already been counted” (or “Verification failed — please try again” for a bot flag).
- c) 3a3. Voter is returned to the showcase catalog. The vote count and leaderboard remain unchanged.
- d) 3a4. Use case ends without recording a vote.